Definition A.2 (L-smoothness [Yang et al., 2022]). A differentiable function $f : \mathbb{R}^d \to \mathbb{R}^k$ is L-smooth, if for any $x_1, x_2 \in \mathbb{R}^d$ and any output dimension $c \in \{1, \dots, k\}$:

$$\frac{\|\nabla_{x_1} f_c(x_1) - \nabla_{x_2} f_c(x_2)\|_2}{\|x_1 - x_2\|_2} \leq L \ .$$

Next we restate one property of L-smooth functions which we will use in our proof of theorem 2.

Proposition A.2 (Bubeck [2015]). *Let f be an L-smooth function on $\mathbb{R}^n$. For any $x, y \in \mathbb{R}^n$ it holds:*

$$|f(y) - f(x) - \langle \nabla_x f(x), y - x \rangle| \leq \frac{L}{2} \|y - x\|_2^2 \ .$$

Proof. From the fundamental theorem of calculus we know that for a differentiable function f it holds that $f(y) - f(x) = \int_x^y \nabla_t f(t) dt$. By substituting $x_t = x + t(y - x)$ we see that $x_0 = x$ and $x_1 = y$ and thus we can write $f(y) - f(x) = \int_0^1 \nabla f(x + t(y - x))^T \cdot (y - x) dt$. This allows the following approximations

$$\begin{aligned}
& |f(y) - f(x) - \langle \nabla_x f(x), y - x \rangle| \\
=& |\int_0^1 \nabla f(x + t(y - x))^T \cdot (y - x) dt - (\nabla_x f(x))^T (y - x)| \\
\leq& \int_0^1 |(\nabla f(x + t(y - x)) - \nabla_x f(x))^T) \cdot (y - x)| dt \\
\overset{\text{Cauchy-Schwarz}}{\leq}& \int_0^1 \|\nabla f(x + t(y - x)) - \nabla_x f(x)\|_2 \cdot \|y - x\|_2 dt \\
\overset{L\text{-smoothness}}{\leq}& L \cdot \|y - x\|_2^2 \cdot \int_0^1 t dt \\
=& \frac{L}{2} \cdot \|y - x\|_2^2 \ .
\end{aligned}$$

$\square$

Theorem A.3 (Sufficient condition for the robustness of an L-smooth stochastic classifier). *Let $f : \mathbb{R}^d \times \Omega^h \to \mathbb{R}^k$ be a stochastic classifier with L-smooth discriminant functions and $f^{\mathcal{A}}$ and $f^{\mathcal{I}}$ be two MC estimates of the prediction. Let $x \in \mathbb{R}^d$ be a data point with label $y \in \{1, \dots, k\}$ and $\arg\max_c f_c^{\mathcal{A}}(x) = \arg\max_c f_c^{\mathcal{I}}(x) = y$, and let $x_{adv} = x + \delta^{\mathcal{A}}$ be an adversarial example computed for solving the minimization problem* (5) *for $f^{\mathcal{A}}$. It holds that $\arg\max_c f_c^{\mathcal{I}}(x + \delta^{\mathcal{A}}) = y$ if*

$$\min_{c \neq y} r_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2 \ ,$$

with

$$r_c^{\mathcal{I}} = \begin{cases} \infty \ , \textit{if} \ \|\nabla_x (f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2} \cdot \|\delta^{\mathcal{A}}\|_2 \leq 0, \\ \frac{f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x)}{\|\nabla_x (f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2} \cdot \|\delta^{\mathcal{A}}\|_2} \ , \textit{else} \end{cases}$$

and

$$\cos(\alpha_c^{\mathcal{I},\mathcal{A}}) = \frac{\langle -\nabla_x (f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x)), \delta^{\mathcal{A}} \rangle}{\|\nabla_x (f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))\|_2 \cdot \|\delta^{\mathcal{A}}\|_2} \ .$$

Proof. For better readability we write $f^{\mathcal{I}}_{y-c}(x) := f^{\mathcal{I}}_y(x) - f^{\mathcal{I}}_c(x)$. Using the result from proposition A.2 and reordering the terms, we get the following lower bound:

$$\begin{aligned}
&f^{\mathcal{I}}_{y-c}(x+\delta^{\mathcal{A}}) \\
&\geq f^{\mathcal{I}}_{y-c}(x) + \langle \nabla_x (f^{\mathcal{I}}_{y-c}(x)), \delta^{\mathcal{A}}\rangle - \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2^2 \\
&= f^{\mathcal{I}}_{y-c}(x) - \langle -\nabla_x (f^{\mathcal{I}}_{y-c}(x)), \delta^{\mathcal{A}}\rangle - \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2^2 \\
&= f^{\mathcal{I}}_{y-c}(x) - \| -\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \|\delta^{\mathcal{A}}\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) - \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2^2 \\
&= f^{\mathcal{I}}_{y-c}(x) - \left(\|\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2 \right)\cdot \|\delta^{\mathcal{A}}\|_2 \ . \qquad (9)
\end{aligned}$$

If eq. (9) is bigger than zero, the attack cannot be successful. Hence,

$$f^{\mathcal{I}}_{y-c}(x) - \left(\|\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2 \right)\cdot \|\delta^{\mathcal{A}}\|_2 \overset{!}{>} 0 \qquad (10)$$

$$f^{\mathcal{I}}_{y-c}(x) > \left(\|\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2 \right)\cdot \|\delta^{\mathcal{A}}\|_2 \ . \qquad (11)$$

Case 1: $\|\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2 > 0$. Transforming eq. (11) leads to

$$\frac{f^{\mathcal{I}}_{y-c}(x)}{\|\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2} > \|\delta^{\mathcal{A}}\|_2 \ .$$

Case 2 : $\|\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2 = 0$. In this case eq. (10) is trivially fulfilled because $f^{\mathcal{I}}_{y-c}(x) > 0$ per definition.

Case 3: $\|\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2 < 0$. For this case we get, that

$$-\frac{f^{\mathcal{I}}_{y-c}(x)}{|\|\nabla_x f^{\mathcal{I}}_{y-c}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2}\cdot \|\delta^{\mathcal{A}}\|_2|} < \|\delta^{\mathcal{A}}\|_2 \ ,$$

which is always guaranteed based on the initial assumption that the benign input was classified correctly, which concludes the proof. □

The following proposition relates to footnote 5 from the main paper. We show, that the interval, in which the expectation of the gradient norm lies, decreases to the true length of μ with reducing the covariance.

Proposition A.4. *Let X be an n-dimensional random vector following a multivariate normal distribution with mean vector μ and diagonal covariance matrix Σ. Then the expectation of $\|X\|_2$ can be upper and lower bounded by*

$$\|\mu\|_2 \leq \mathbb{E}[\|X\|_2] \leq \sqrt{\|\mu\|_2^2 + tr\,(\Sigma)} \ .$$

Proof. We first look at the lower bound which by convexity of the norm and Jensen inequality can be derived via

$$\mathbb{E}[\|X\|_2] \leq \|\mathbb{E}[X]\|_2 = \|\mu\|_2 \ .$$

Let $X' \sim \mathcal{N}(0, \mathbf{1}_n)$, with $\mathbf{1}_n$ an $n \times n$-dimensional unit matrix. With Jensen inequality and concavity of the square-root function we derive the upper bound

$$\begin{aligned}
&\mathbb{E}[\|X\|_2] = \mathbb{E}\left[\|\mu + X'\cdot\Sigma^{1/2}\|_2\right] \\
&\leq \left(\mathbb{E}\left[\left\|\mu + X'\cdot\Sigma^{1/2}\right\|_2^2\right]\right)^{1/2} \\
&= \left(\|\mu\|_2^2 + 2\mu^T\Sigma^{1/2}\mathbb{E}[X'] + \mathbb{E}\left[X'^T\Sigma^{1/2^T}\Sigma^{1/2}X'\right]\right)^{1/2} \\
&= \sqrt{\|\mu\|_2^2 + tr\,(\Sigma)} \ .
\end{aligned}$$

$\square$

B Additional information on datasets, models and training

In the following we describe additional details on the datasets, models and training procedures which were not stated in the main paper due to space restrictions. Additionally, please find the code for the results in the main paper attached in the supplementary material.

B.1 Datasets

We used three well know datasets: FashionMNIST [Xiao et al., 2017], CIFAR10 and CIFAR100 [Krizhevsky et al., b], which consist out of 60,000 training and 10,000 test images of dimension 28×28 or $32 \times 32 \times 3$ in case of CIFAR where each image is uniquely associated to one out of 10 or 100 possible labels. We took all datasets from the torchvision package with the predefined training and test split.

B.2 Models trained on FashionMNIST

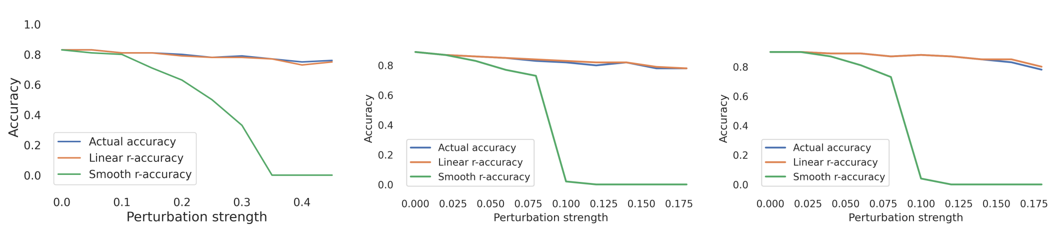
For training the BNN and IM we used the exact same hyperparameters. First, we assumed a standard normal prior decomposed as matrix variate normal distributions for the BNN and approximated the posterior distribution via maximizing the evidence lower bound (ELBO). For the IM we added a Kullback-Leibler distance from the trained parameter distribution to a standard matrix variate normal distribution as a regularization term. For both models we used a batch size of 100 and trained for 50 epochs with Adam [Kingma and Ba, 2015] and an initial learning rate of 0.001. To leverage the difference between IM and BNN we used 5 samples to approximate the expectation in the ELBO/IM-objective. We used the same learning rate, batch size, optimizer and amount of epochs for training the stochastic input networks. During a forward pass in training we created and used five noisy versions of each input, where the noise was drawn from a centered Gaussian distribution with variance 0.05 or 0.1 and the average prediction was fed into the cross-entropy loss.

B.3 Models trained on CIFAR10

As stated in the main part, we used the wide ResNet [Zagoruyko and Komodakis, 2017] of depth 28 and widening factor 10 provided by `https://github.com/meliketoy/wide-resnet.pytorch` with dropout probabilities 0.3 and 0.6 and also used the learning hyperparameters provided with the code which are: training for 200 epochs with batch size 100, stochastic gradient descent as optimizer with momentum 0.9, weight decay 5e-4 and a scheduled learning rate decreasing from an initial 0.1 for epoch 0-60 to 0.02 for 60-120 and lastly 0.004 for epochs 120-200.

C Additional experimental results

In this section we present the results which were not shown in the main part due to space restrictions.



(a) IM (b) dr 0.3 (c) dr 0.6

Figure 7: Adversarial accuracy of the a) smoothed IM on FashionMNIST and b),c) smoothed ResNet with dropout probability 0.3 and 0.6 on CIFAR10 vs percentage of images for which $\min_c r_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2$ (smooth) and $\min_c \tilde{r}_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2$ (linear) for 100 images from the respective test sets. Attacks were conducted on the smoothed classifier with using 10 attack samples.